

VAN / SURNAME: MEMO

VOORLETTERS / INITIALS: _____ REG.No.: _____

HANDTEKENING / SIGNATURE: _____

KURSUS / COURSE: FSK 176

UNIVERSITEIT VAN PRETORIA
THE UNIVERSITY OF PRETORIA

FSK 176

SEMESTERTOETS / SEMESTER TEST

17 AUGUSTUS / AUGUST 2011

Totale Punte / Total Marks: 100

Tyd / Time: 2 ure / hours

- Beantwoord alle vrae. / Answer all questions.
- Programmeerbare sakrekenaars mag **NIE** gebruik word **NIE**
- Programmable pocket calculators may **NOT** be used.
- Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.
All test and exam regulations of the University of Pretoria are applicable.

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FSK 176: SEMESTERTOETS / SEMESTER TEST
17 AUGUSTUS / AUGUST 2011

VAN EN VOORLETTERS: _____

SURNAME AND INITIALS: _____

STUDENTE NR. _____

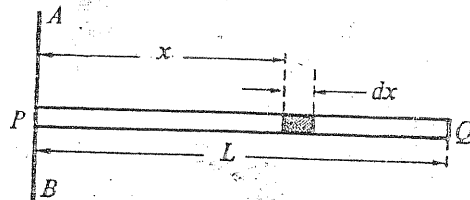
STUDENT NUMBER: _____

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SIGNATURE: _____

VRAAG / QUESTION 1

Die skets toon 'n dun reguit homogene staaf met massa M kilogram en lengte L meter. Die massa per eenheidslengte word gegee deur $\mu = M/L \text{ kgm}^{-1}$. AB is 'n as wat loodreg is op PQ en dx is 'n infinitesimale lengte-element van die staaf op afstand x vanaf AB . Die massa van hierdie element word nou gegee deur $dm = \mu dx = (M/L)dx$. Die hoeveelheid $dI = x^2 dm$ staan bekend as die traagheidsmoment van die element dx om die as AB .

The sketch shows a thin straight homogeneous rod with length L meters and mass M kilograms. The mass per unit length is given by $\mu = M/L \text{ kgm}^{-1}$. AB is an axis perpendicular to the rod PQ , and dx is an infinitesimal element of length at position x from AB . The mass of this element is given by $dm = \mu dx = (M/L)dx$. The quantity $dI = x^2 dm$ is known as the moment of inertia of the element about the axis AB .



- a) Bereken die traagheidsmoment I van die hele staaf om AB .
Calculate the moment of inertia I of the entire rod about axis AB

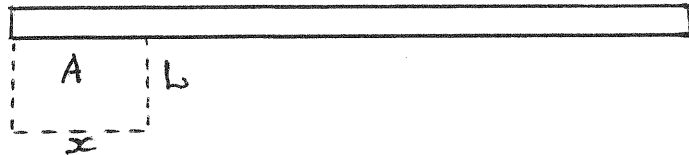
$$\begin{aligned}
 dI &= x^2 dm \\
 &= x^2 \left(\frac{M}{L} dx \right) \checkmark \\
 \int_0^L dI &= \frac{M}{L} \int_0^L x^2 dx \checkmark \\
 &= \frac{M}{L} \left[\frac{1}{3} x^3 \right]_0^L \\
 &= \frac{1}{3} ML^2 \Rightarrow \checkmark \quad (3)
 \end{aligned}$$

- b) Bereken ook die traagheidsmoment van die staaf om 'n as ewewydig aan AB en deur die middelpunt van PQ .
Calculate the moment of inertia of the rod about an axis parallel to AB and through the centre of the rod PQ .

$$\begin{aligned}
 \int_0^L dI &= \frac{M}{L} \int_{-L/2}^{+L/2} x^2 dx \checkmark \\
 I &= \frac{M}{L} \left[\frac{1}{3} x^3 \right]_{-L/2}^{+L/2} \checkmark \\
 &= \frac{1}{12} ML^2 \Rightarrow \checkmark \quad (3)
 \end{aligned}$$

- c) Daar bestaan 'n lang reguit klipmuur op 'n plaas. Dit is ongeveer 200 m lank. Die boer besit genoeg kampmateriaal om 'n draadheining van 100 m lank te maak. Hy gebruik die bestaande klipmuur as een sy en voltooi die ander drie sye van 'n reghoekige kamp met sy beskikbare materiaal. Bereken die afmetings van die kamp wat so gebou word dat die oppervlakte daarvan 'n maksimum sal wees.

An existing stone wall on a farm is about 200 m long. The farmer has enough material to erect a fence of length 100 m. He wishes to use a portion of the existing wall as the one side of a rectangular pen for his cattle and the other three sides are to be completed by using the available fencing material. Calculate the dimensions and the area of the pen which is constructed in this way and will have a maximum area.



- (i) Gee 'n uitdrukking vir L in terme van x. / Give an expression for L in terms of x.

$$2L + x = 100 \quad \checkmark$$

$$\therefore L = 50 - \frac{1}{2}x \quad \checkmark$$

(2)

- (ii) Herlei nou 'n uitdrukking vir A in terme van x. / Derive an expression for A in terms of x

$$A = xL \quad \checkmark$$

$$= x(50 - \frac{1}{2}x) = 50x - \frac{1}{2}x^2 \quad \checkmark$$

(2)

- (iii) Toon aan dat $x = 50$ m. / Show that $x = 50$ m.

$$\frac{dA}{dx} = 50 - x = 0 \quad \checkmark$$

$$\therefore x = 50 \quad \checkmark$$

(2)

- (iv) Bepaal die waarde van L. / Determine the value of L

$$L = 50 - \frac{1}{2}x$$

$$= 50 - \frac{1}{2}(50) = 25 \quad \checkmark$$

(1)

- (v) Bewys dat die oppervlakte A wel 'n maksimum is. / Proof that area A is really a maximum,

$$\frac{d^2A}{dx^2} = -1 < 0 \quad \checkmark \quad \therefore \text{max}$$

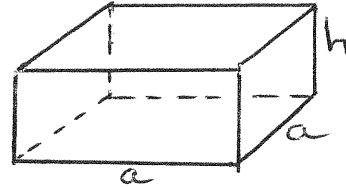
(2)

VRAAG/QUESTION 2

'n Vierkantige tenk, wat toe is by sy bokant, word van metaalplate gemaak. Die kapasiteit van die tenk is $V \text{ m}^3$. Die vierkantige basis se sylengtes is a en die diepte is h .

A square tank is closed on the top side and is manufactured from sheet iron. The capacity of the tank is $V \text{ m}^3$. The square base has side lengths a and the depth of the tank is h .

- (i) Gee 'n uitdrukking vir die hoogte h in terme van die volume V van die tenk.
Give an expression for the height h in terms of the volume V of the tank.



$$V = a^2 h \checkmark$$

$$\therefore h = \frac{V}{a^2} \checkmark$$

(2)

- (ii) Indien die volume $V = 8 \text{ m}^3$, toon aan dat die oppervlakte van al die plate is $A = 32a^{-1} + 2a^2$.
If the volume $V = 8 \text{ m}^3$, the area of all the sheets is $A = 32a^{-1} + 2a^2$.

$$A = 4(ah) + 2a^2 \checkmark$$

$$\text{But } h = \frac{V}{a^2}$$

$$h = \frac{8}{a^2} \checkmark$$

$$A = 4a\left(\frac{8}{a^2}\right) + 2a^2 \checkmark$$

$$= \frac{32}{a} + 2a^2 \Rightarrow$$

(3)

- (iii) Bereken die sylengtes a sodat die minste metaal moontlik gebruik sal word.

Determine the lengths a so that the least amount of sheet iron may be used.

$$\frac{dA}{da} = \frac{d}{da} \left[\frac{32}{a} + 2a^2 \right] \checkmark$$

$$= -32a^{-2} + 4a$$

$$= 0 \checkmark$$

$$\therefore -32a^{-2} + 4a = 0$$

$$32a^{-2} = 4a \checkmark$$

$$a^2(a) = \frac{32}{4} = 8$$

$$a^3 = 8$$

$$a = 2 \Rightarrow \checkmark$$

(4)

(iv) Wat is die hoogte h van die tenk?

What is the depth h of the tank?

$$h = \frac{V}{a^2} \\ = 8/(2)^2 \quad \checkmark$$

$$\therefore h = 2 \Rightarrow \checkmark$$

(2)

[11]

VRAAG/QUESTION 3

- (a) Definieer spesifieke warmte c van 'n liggaam en gee die eenheid waarin joule (J) voorkom.

Define specific heat c of an object and give the unit which contains joule (J).

Specific heat c is the amount of
energy $\checkmark$ necessary to increase the
temp. of 1 kg $\checkmark$ of the material
by 1°C $\checkmark$
J/kg °C $\checkmark$

(4)

- (b) Die interne energie E_{int} van 'n sisteem is onafhanklik van die toestand van die sisteem, soos bepaal deur druk P , volume V en temperatuur T . Gee die eerste wet van termodinamika:

The internal energy E_{int} of a system is independent of the state of a system, as determined by pressure P , volume V and temperature T . Give the first law of thermodynamics:

$$dE_{\text{in}} = dQ - dw \quad \checkmark$$

(1)

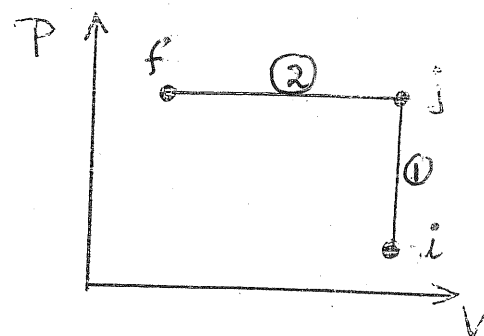
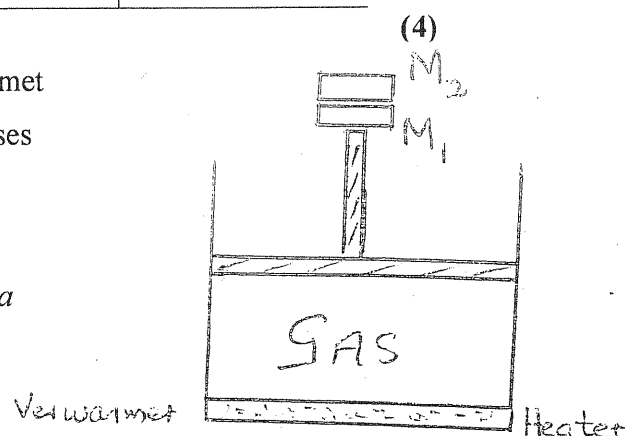
(c) Voltooi die tabel vir die volgende prosesse:

Complete the table for the following processes:

Proses Process	Voorwaarde / Condition ($Q = 0$ of/ or $W = 0$)	Interne energie / Internal energy $\Delta E_{\text{int}} =$
adiabaties adiabatic	$Q = 0$	$-W$
Isovolumetrics Isovolumetric	$W = 0$	Q

(d) 'n Gas is ingesluit in 'n geïsoleerde silinder met 'n beweegbare suier. 'n Termodinamiese proses ($i \rightarrow f$) word aangetoon.

A gas is confined to an isolated cylinder with a movable piston. A thermodynamic process ($i \rightarrow f$) is shown.



Gebruik die volgende simbole en voltooi die tabel:

Use the following symbols to complete the table:

- c = konstant / constant
- > = neem toe / increase
- < = neem af / decrease
- 0 = 0

Stap / Step	P	V	T	Warmte / Heat Q Arbeid / Work W (+ of / or -)
① $i \rightarrow j$	>	c	>	+
② $j \rightarrow f$	c	<	<	-

(8)

(e) Verwyder die massa M_2 en toon aan dat die arbeid wat verrig word gegee word deur:

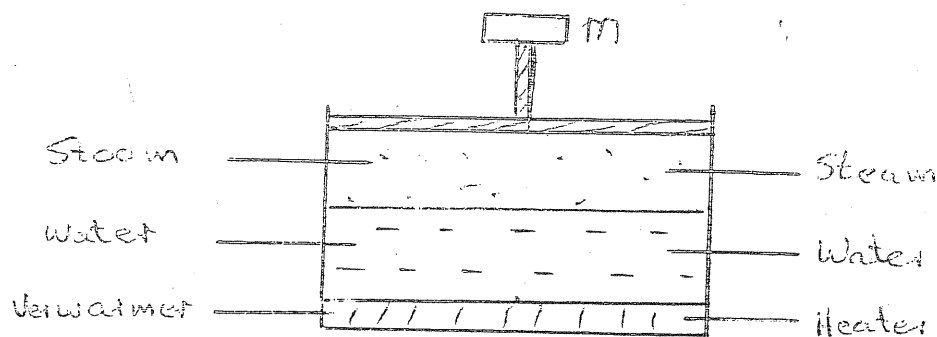
Remove the mass M_2 and show that the work done is given by:

$$\begin{aligned}
 W &= \int_{V_i}^{V_f} p dV \\
 dW &= \vec{F} \cdot d\vec{s} \quad \checkmark \\
 &= F ds \cos 0^\circ \\
 &= (pA) ds \quad \checkmark \\
 &= p (A ds) \quad \checkmark \\
 &= p dV \quad \checkmark \\
 W &= \int_{V_i}^{V_f} p dV
 \end{aligned}$$

(4)

- (f) Wanneer water by atmosferiese druk ($1,01 \times 10^5 \text{ Pa}$) gekook word, word 1000 g water by 100° C omgesit in stoom by 100° C . Die volume ($1 \times 10^{-3} \text{ m}^3$) water as vloeistof word omgesit na $1,6842 \text{ m}^3$ stoom.

Let 1000 g of liquid water at 100° C be converted to steam at 100° C by boiling at atmospheric pressure ($1,01 \times 10^5 \text{ Pa}$). The volume of liquid water ($1 \times 10^{-3} \text{ m}^3$) changes to steam with a volume of $1,6842 \text{ m}^3$.



- (i) Hoeveel arbeid (in kJ) word deur die sisteem verrig tydens hierdie proses?

How much work (in kJ) is done by the system during this process?

$$\begin{aligned}
 W &= \int_{V_i}^{V_f} p \, dV \\
 &= p \int_{V_i}^{V_f} dV \quad \checkmark \\
 &= p(V_f - V_i) \quad \checkmark \\
 &= (1,01 \times 10^5) (1,6842 - 1) \times 10^{-3} \quad \checkmark \\
 &= 170 \text{ kJ} \Rightarrow \checkmark
 \end{aligned}$$

(4)

- (ii) Wanneer vloeibare water (by 100° C) verander word in stoom (by 100° C) sal die warmte Q slegs 'n faseverandering te weeg bring, maar geen verandering in die temperatuur nie. Indien die verdampingswarmte van water $L = 2260 \text{ kJ/kg}$ is, hoeveel warmte Q word in hierdie proses oorgedra?

When liquid water (at 100° C) is changed to steam (at 100° C) the

heat Q causes only a phase change and not a change in temperature.

If the heat of vaporization of water $L = 2260 \text{ kJ/kg}$, how much heat Q is transferred during this process?

$$\begin{aligned}
 Q &= mL \quad \checkmark \\
 &= 1 (2260) \\
 &= 2260 \text{ kJ} \Rightarrow \checkmark
 \end{aligned}$$

(2)

- (iii) Bepaal nou die verandering in die sisteem se interne energie ΔE_{int} .

Now determine the change in the system's internal energy ΔE_{int} .

$$\begin{aligned}
 \Delta E_{\text{int}} &= Q - W \quad \checkmark \\
 &= 2260 - 170 \\
 &= 2090 \text{ kJ} \Rightarrow \checkmark
 \end{aligned}$$

(2)

- (iv) Die interne energie (ΔE_{int}) word aangewend om die watermolekules te skei (van vloeistof na stoom). Watter % van die warmte ($Q = 2260 \text{ kJ}$) word gebruik as arbeid om die atmosfeer terug te druk?

The internal energy (ΔE_{int}) goes into separating the water molecules (from liquid to steam). What % of the heat ($Q = 2260 \text{ kJ}$) goes into the work done in pushing back the atmosphere?

$$\frac{170}{2260} \times 100\% = 7,5\%$$

(2)
[31]

VRAAG/QUESTION 4

- (a) Definieer warmtekapasiteit C van 'n voorwerp en gee sy eenheid.

Define heat capacity C of an object and give its unit.

Heat capacity (C) is the amount of energy necessary to increase the temp. of the whole object [hele voorwerp] with 1°C (or 1K)

$$\text{J/K or J/}^\circ\text{C}$$

(4)

- (b) In die uitdrukking $\Delta Q = mc \Delta T$, wat word c genoem en gee sy eenheid.

In the expression $\Delta Q = mc \Delta T$, what is c called and also give its unit.

$c = \text{specific heat}$ [spesifieke warmte]
 J/kg K (or $\text{J/kg}^\circ\text{C}$)

(2)

- (c) Die spesifieke warmtekapasiteit van water (of tee) is $4186 \text{ J/kg }^\circ\text{C}$. Nou word 200 cm^3 tee, wat by 95°C is, in 'n glaskoppie van 150 gram gegooi wat aanvanklik by 'n temperatuur van 25°C is. Die spesifieke warmte van glas is $840 \text{ J/kg }^\circ\text{C}$.

The specific heat of water (or tea) is $4186 \text{ J/kg }^\circ\text{C}$. Now 200 cm^3 of tea at 95°C is poured into a glass cup of 150 gram initially at 25°C . The specific heat of glass is $840 \text{ J/kg }^\circ\text{C}$.

- (i) Bepaal die massa m van die water (tee) in die koppie indien die digtheid van water $1 \times 10^3 \text{ kg/m}^3$ is.

Determine the mass m of the water (tea) in the cup if the density of water is $1 \times 10^3 \text{ kg/m}^3$ is.

$$\begin{aligned} m &= \rho V \quad \checkmark \\ &= (1 \times 10^3) (200 \times 10^{-6}) \quad \checkmark \\ &= 0,2 \text{ kg} \Rightarrow \checkmark \end{aligned} \quad (3)$$

- (ii) Indien die finale temperatuur T is, gee 'n uitdrukking vir die warmte Q_{tea} wat die tee verloor het.

If the final temperature is T , give an expression for the heat Q_{tea} lost by the tea.

$$\begin{aligned} Q_{\text{tea}} &= m_{\text{tea}} c_{\text{tea}} (95 - T) \\ &= 0,2 (4186) (95 - T) \quad \checkmark \end{aligned} \quad (2)$$

- (iii) Die warmte Q_{cup} wat deur die koppie opgeneem is word gegee deur:
The heat Q_{cup} gained by the cup is:

$$\begin{aligned} Q_{\text{cup}} &= m_{\text{cup}} c_{\text{cup}} (T - 25) \\ &= 0,15 (840) (T - 25) \quad \checkmark \end{aligned} \quad (2)$$

- (iv) Bereken met hoeveel die temperatuur van die tee afgeneem het nadat ewewig bereik is.

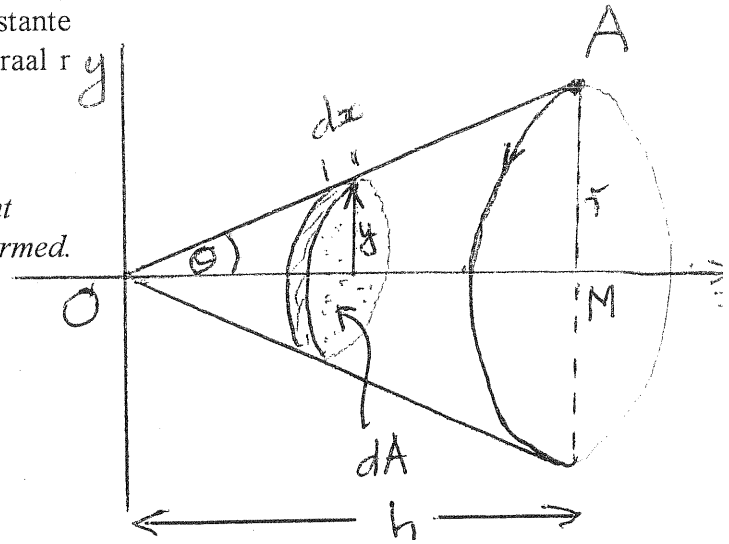
Determine by how much the temperature of the tea dropped after equilibrium was reached.

$$\begin{aligned} 0,2 (4186) (95 - T) &= 0,15 (840) (T - 25) \quad \checkmark \\ 79534 - 837T &= 126T - 3150 \quad \checkmark \\ \therefore 963T &= 82684 \quad \checkmark \\ T &\approx 86 \quad \checkmark \\ \text{Drop} &\approx 95 - 86 \approx 9^\circ\text{C} \Rightarrow \checkmark \end{aligned} \quad (4)$$

VRAAG/QUESTION 5

- (a) Wanneer 'n reguitlyn $y = mx$ (di OA) geroteer word om die x-as met 'n konstante hoek θ , dan word 'n sirkelkeël met straal r gevorm.

When a straight line $y = mx$ (i.e OA) is rotated about the x-axis with a constant angle θ , then a cone with radius r is formed.



- (i) Wat is die oppervlakte dA van die segment met straal y .

What is the area dA of the segment with radius y .

$$dA = \pi y^2$$

(1)

- (ii) Wat is die volume dV van die segment met dikte dx .

What is the volume dV of the segment with thickness dx .

$$dV = dA dx$$

$$= \pi y^2 dx$$

(1)

- (iii) As die hoogte van die sirkelkeël h is, dan is die helling m van lyn OA:

If the height of the cone is h , then the gradient m of line OA is:

$$m = \frac{r}{h}$$

(1)

(iv) Bereken die totale volume V (tussen $x = 0$ en $x = h$) van die sirkelkeël wat deur die rotasie van die lyn $y = mx$ gevorm word.

Determine the total volume V (between $x = 0$ and $x = h$) of the cone which was formed by the rotation of the line $y = mx$.

$$\begin{aligned}
 V &= \int_0^h dV \\
 &= \int_0^h \pi y^2 dx \checkmark \\
 &= \pi \int_0^h (mx)^2 dx \\
 &= \pi m^2 \left[\frac{1}{3} x^3 \right]_0^h \checkmark \\
 &= \frac{1}{3} \pi m^2 h^3 \checkmark \\
 &= \frac{1}{3} \pi \left(\frac{r}{h} \right)^2 h^3 \\
 &= \frac{1}{3} \pi r^2 h \Rightarrow \checkmark
 \end{aligned}$$

(6)

- (b) 'n Reënmeter is 'n sirkelkeël met gegewens in die skets en sy volume (soos in iv) is:
A rain gauge is conical with dimensions as given in the figure. The volume is (as given in (iv)):

$$V = \frac{1}{3} \pi r^2 h$$

Uit die driehoeke TBC en TAD, elimineer r uit hierdie vergelyking en toon dat

$$V = \frac{1}{3} \pi \frac{R^2}{H^2} h^3$$

From the triangles TBC and TAD, eliminate r from this expression and show that

$$V = \frac{1}{3} \pi \frac{R^2}{H^2} h^3$$

$$V = \frac{1}{3} \pi \left(\frac{h}{H} R \right)^2 h \checkmark$$

$$V = \frac{1}{3} \pi \frac{R^2}{H^2} h^3$$

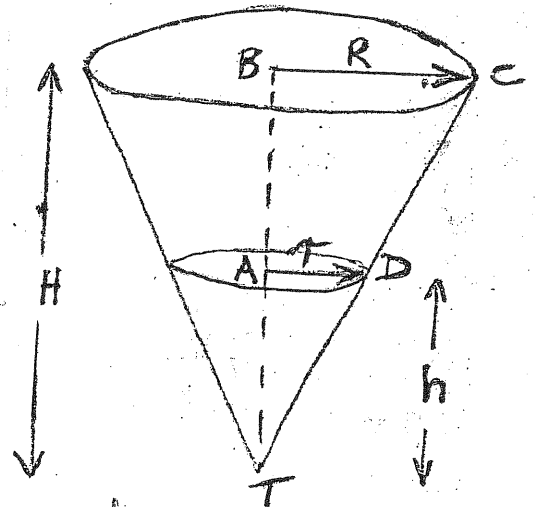
(2)

- (c) Uit bostaande, skryf h as 'n funksie van die volume V .
From above, write h as a function of its volume V .

$$h^3 = \frac{3V}{\pi} \frac{H^2}{R^2}$$

$$\therefore h = \left(\frac{3}{\pi} \frac{H^2}{R^2} V \right)^{1/3} \checkmark$$

(2)



- (d) Bereken nou die tempo waarmee die hoogte verander as die volume toeneem $\left(\frac{dh}{dV}\right)$.

Calculate the rate at which the height changes as the volume increases $\left(\frac{dh}{dV}\right)$.

$$\frac{dh}{dV} = \left(\frac{3}{\pi} \frac{H^2}{R^2}\right)^{1/3} \frac{d}{dV} V^{1/3} \quad \checkmark$$

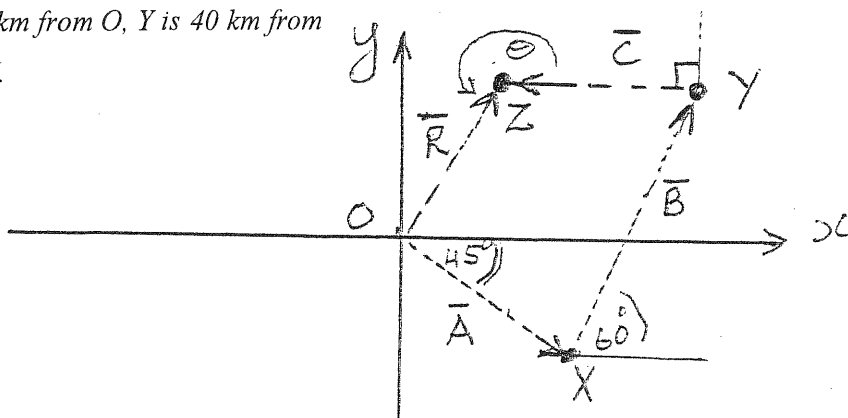
$$= \left(\frac{3}{\pi} \frac{H^2}{R^2}\right)^{1/3} \frac{1}{2} V^{-2/3} \quad \checkmark$$

(2)
[15]

VRAAG/QUESTION 6

- (a) 'n Persoon stap teen 3,5 km/h. In die rigting soos aangetoon is X 25 km vanaf O, Y is 40 km vanaf X en Z is 20 km vanaf Y.

A person walks at 3,5 km/h. In the directions as indicated, point X is 25 km from O, Y is 40 km from X and Z is 20 km from Y.



- (i) Bepaal die tyd (in uur) wat elke been van die staptog geneem het. (Gee u antwoorde tot 1 desimale plek).

Determine the time (in hour) it took to walk each leg of the path. (Give your answer with 1 decimal place).

$$t_A = \frac{s}{v} = \frac{25}{3,5} = 7,1 \text{ h} \Rightarrow$$

$$t_B = \frac{40}{3,5} = 11,4 \text{ h} \Rightarrow$$

$$t_C = \frac{20}{3,5} = 5,7 \text{ h} \Rightarrow$$

(3)

- (ii) Gee die vektore $\vec{A}$, $\vec{B}$ en $\vec{C}$ in die eenheidsvektor-notasie.

Give the vectors $\vec{A}$, $\vec{B}$ en $\vec{C}$ in the unit vector notation.

$$\begin{aligned}\vec{A} &= 25 \cos(-45) \hat{i} + 25 \sin(-45) \hat{j} \checkmark \\ &= 17,68 \hat{i} - 17,68 \hat{j} \\ \vec{B} &= 20 \hat{i} + 34,64 \hat{j} \Rightarrow \checkmark \\ \vec{C} &= -20 \hat{i} \Rightarrow \checkmark\end{aligned}\quad (3)$$

- (iii) Gee die resultante verplasing $\vec{R}$ in terme van eenheidsvektore.

Give the resultant displacement $\vec{R}$ in terms of unit vectors.

$$\begin{aligned}\vec{R} &= \vec{A} + \vec{B} + \vec{C} \\ &= (17,68 \hat{i} - 17,68 \hat{j}) + (20 \hat{i} + 34,64 \hat{j}) + (-20 \hat{i}) \\ &= 17,7 \hat{i} + 16,9 \hat{j} \Rightarrow \checkmark\end{aligned}\quad (1)$$

- (iv) By punt X het die persoon vir 20 min gerus, en by punt Y was dit 40 min. Wat is die gemiddelde spoed $\langle v \rangle$ van die persoon?

At point X the person rested for 20 min, and at point Y it was 40 min. What is the average speed $\langle v \rangle$ of the person?

$$\begin{aligned}\langle v \rangle &= \frac{\Delta x}{\Delta t} = \frac{25 + 40 + 20}{(7,1 + 11,4 + 5,7 + 1)} \checkmark \\ &= \frac{85}{25,2} \\ &= 3,37 \text{ km/h} \Rightarrow \checkmark\end{aligned}\quad (2)$$

Wat was die gemiddelde snelheid $\vec{v}$ vir die hele tog?

What was the average velocity $\vec{v}$ for the whole trip?

$$\begin{aligned}\vec{v} &= \vec{R} / \Delta t \\ &= \frac{17,7}{25,2} \hat{i} + \frac{16,9}{25,2} \hat{j} \checkmark \\ &= 0,7 \hat{i} + 0,67 \hat{j} \Rightarrow \checkmark\end{aligned}\quad (2)$$